2 Subject content

The Pearson Edexcel Level 1/Level 2 GCSE (9–1) in Statistics ensures that students develop the confidence and competence with statistical techniques to enable them to apply those techniques flexibly to solve statistical problems through a practical programme of study, with the expectation that:

- all students (both tiers) will develop confidence and competence with the content identified by the standard type
- all students will be assessed on the content identified by the standard type and the underlined type, and this content will be in both foundation tier and higher tier papers. The more highly-attaining students will develop confidence and competence with all this content
- only the more highly-attaining students (Higher tier only) will be assessed on the content identified by **bold** type; the highest-attaining students will develop confidence and competence with this content
- the distinction between standard, underlined and **bold** type applies to the content statements only, and not to assessment objectives or to the mathematical formulae in *Appendix 1*
- all students will develop an appreciation that different approaches, including the use of technology, may be appropriate at each stage of the statistical enquiry cycle, and that statistical conclusions are developed through an iterative process of retesting and refinement
- all students have the opportunity to apply statistical techniques within the framework of the **statistical enquiry cycle** using real data from authentic contexts.

Please see *Appendix 3* for details of formulae in both GCSE Mathematics and Statistics which will not be given in the assessments, and *Appendix 2* for details of prior knowledge.

Qualification aims and objectives

The aims and objectives of this qualification are to enable students to develop statistical fluency and understanding through:

- the use of statistical techniques in a variety of authentic investigations, using real-world data in contexts such as, but not limited to, populations, climate, sales etc.
- identifying trends through carrying out appropriate calculations and data visualisation techniques
- the application of statistical techniques across the curriculum, in subjects such as the sciences, social sciences, computing, geography, business and economics, and outside the classroom in the world in general
- critically evaluating data, calculations and evaluations that would be commonly encountered in their studies and in everyday life
- understanding how technology has enabled the collection, visualisation and analysis of large quantities of data to inform decision-making processes in public, commercial and academic sectors, including how technology can be used to generate diagrams and visualisations to represent data
- understand ways that data can be organised, processed and presented, including statistical measures to compare data, understanding the advantages of using technology to automate processing
- applying appropriate mathematical and statistical formulae, and building on prior knowledge.

Statistical enquiry cycle

The order of the content, for each tier, follows the order of the **statistical enquiry cycle**. It is important that practical investigations are part of a programme of study so that students have the opportunity to understand that different approaches, including the use of technology, may be appropriate at each stage of the statistical enquiry cycle, and that statistical conclusions are developed through an iterative process of testing and refinement.

1. Through using the statistical enquiry cycle students need to understand the **importance of initial planning** when designing a line of enquiry or investigation, including:
   - defining a question or hypothesis (or hypotheses) to investigate
   - deciding what data to collect and how to collect and record it giving reasons
   - developing a strategy for how to process and represent data giving reasons.
2. Through using the statistical enquiry cycle students need to be able to recognise the **constraints involved in sourcing appropriate data**, including:
   - when designing collection methods for primary data
   - when researching sources for secondary data, including from reference publications, the internet and the media
   - through appreciating the importance of acknowledging sources
   - by recognising where issues of sensitivity may influence data availability.
3. Through using the statistical enquiry cycle students need to understand **ways that data can be processed and presented**, including:
   - organising and processing data, including an **understanding of how technology can be used**
   - generating diagrams and visualisations to represent the data, including an **understanding of outputs generated by appropriate technology**
   - generating statistical measures to compare data, **understanding the advantages of using technology to automate processing.**
4. Through using the statistical enquiry cycle students need to understand that **results must be interpreted with reference to the context of the problem**, including:
   - analysing/interpreting the diagrams and calculations/measures
   - reaching conclusions that relate to the questions and hypotheses addressed
   - making inferences and/or predictions
   - discussing the reliability of findings.
5. Through using the statistical enquiry cycle students should show an understanding of the importance of **clear and concise communication** of findings and key ideas, and an **awareness of target audience**. They should also understand the importance of **evaluating statistical work**, including:
   - identifying weaknesses in approach or representation
   - suggesting improvements to processes or the presentation
   - refining the processes to elicit further clarification of the initial hypothesis.

Any given question may assess one stage of the *statistical enquiry cycle* or more than one stage of the statistical enquiry cycle. For example, please see the *Pearson Edexcel*

Foundation tier content

A variety of contexts, including real-life data, will be used. (No detailed knowledge of those contexts will be expected.) Each item of content has a code, for example **1a.01**, where **1** is 'The collection of data', **a** is 'Planning' and **01** is the numerical order of the content item.

What students need to learn:	Guidance